

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Multimeter, Part Number 3164489

Additional Service Items

- No additional service items required.

General Information

The transmission gear oil pressure sensor is used to measure the pressure of the transmission oil and is located on the marine gear oil cooler.

Note : The content of this procedure is for the Cummins Inc.-supplied sensor. The sensor may be supplied by the original equipment manufacturer (OEM), therefore the Cummins® service tools listed in this procedure may not fit. See equipment manufacturer service information.

Initial Check

Use the ED-4 to monitor the value of the transmission pressure sensor with the key in the ON position and the engine off.

The gear oil pressure displayed on the ED-4 data screen is in absolute atmospheric pressure at the current elevation when the engine is **not** running.

The transmission pressure sensor value should meet the following specification.

- On the ED-4 data screen, the transmission pressure sensor should read between 47.6 kPa [6.9 psi] and 105.5 kPa [15.3 psi] at ambient conditions with lower values corresponding to exposure to higher elevations.

Use the following procedure to troubleshoot sensor issues using the ED-4. Refer to Procedure 015-035 in Section 15 (</qs3/pubsys2/xml/en/procedures/513/513-015-035.html>). If the pressure value is out of specification, replace the pressure sensor.

Preparatory Steps

⚠ WARNING ⚠

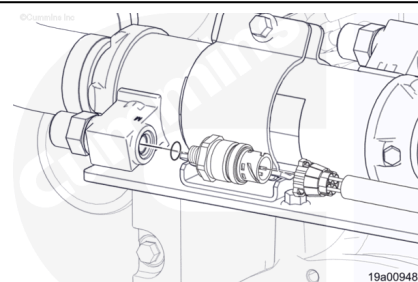
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Clean the area around the sensor.
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSB6.7 CM2250 Service Manual, Bulletin 4310611. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/154/154-008-041-tr.html>)
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSL9 M CM2250 L106 Service Manual, Bulletin 4358343. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/263/263-008-041-tr.html>)
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSC8.3 CM850 Troubleshooting and Repair Manual, Bulletin 4021418. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-041-tr.html>)
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSM11 CM570 Service Manual, Bulletin 3666322. Refer to Procedure 008-084 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-084-tr.html>)

Remove

Disconnect the sensor connector from the harness.

Remove the sensor from the marine gear oil cooler.



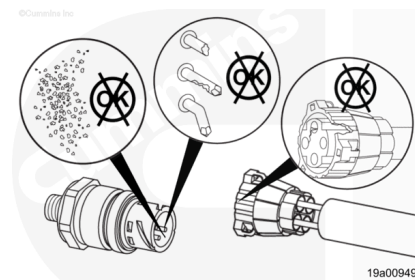
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Clean and Inspect for Reuse

Inspect the harness connector and sensor for the following:

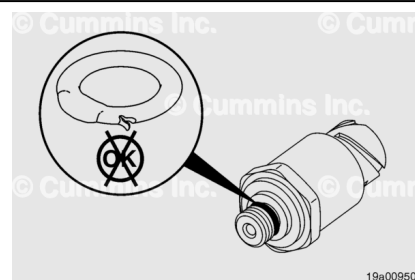
- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins
- Corroded, bent, broken, pushed back, or expanded pins
- Chipped, cracked, extruded, or damaged sensor.

Repair or replace parts as necessary.



Inspect the sensor for the following:

- Swollen o-ring
- Nicks or cuts in or on the o-ring.



Install

Check to be sure the sensor has an o-ring installed.

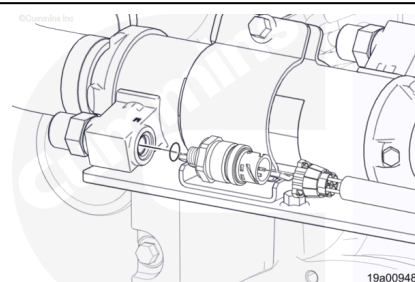
Lubricate the o-ring with clean engine oil before installation.

Install the sensor into the marine gear oil cooler.

Tighten the sensor.

Torque Value: $15 \pm 3 \text{ n}\cdot\text{m}$ [$133 \pm 27 \text{ in}\cdot\text{lb}$]

Connect the harness connector to the sensor.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Fill the marine gear oil cooler. Use the following procedure in the Marine QSB6.7 CM2250 Service Manual, Bulletin 4310611. Refer to Procedure 008-041 in Section 8.
(/qs3/pubsys2/xml/en/procedures/154/154-008-041-tr.html)
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(/qs3/pubsys2/xml/en/procedures/35/35-008-084-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.
- Perform system test to verify proper function. Refer to Procedure 015-047 in Section 15.
(/qs3/pubsys2/xml/en/procedures/513/513-015-047.html)